WhyLab: A Causal Safety Monitoring Framework for Stable Self-Improving Agents

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Abstract

When do self-improving AI agents fail, and how can failures be predicted before deployment? We study *cognitive policy oscillation*—strategy degradation caused by hallucinated or spuriously correlated feedback—and provide the **first systematic characterization** of the conditions under which it occurs.

Through a phase-diagram analysis spanning 384 conditions (8 noise rates $\times$ 8 step sizes $\times$ 3 drift regimes $\times$ 10 seeds), we identify a **sharp instability boundary**: oscillation emerges at hallucination rates as low as $h{=}0.05$ and intensifies with step size and environmental non-stationarity. We then present **WhyLab**, a conditional causal audit framework that activates *only in the unstable regime*: **(C1)** an information-theoretic drift index; **(C2)** a sensitivity filter combining E-values and partial R^2 bounds; and **(C3)** a Lyapunov-bounded adaptive damping controller.

Four key findings emerge: **(1)** In the unstable regime, WhyLab reduces oscillation by 48% across 56/64 tested conditions, with independent contributions from each component (C3: -49% energy, C2: -76% oscillation, C1: -16% detection delay). **(2)** On an adversarial LLM benchmark (E7v2), the audit reduces regressions by 44% on Gemini 2.0 Flash ($p{=}0.014$); an adaptive threshold variant eliminates accuracy cost on near-ceiling models. **(3)** The phase boundary transfers to real tasks: on 2,430 SWE-bench episodes, the audit reduces oscillation by 13–26% at high retry counts; across three model families (960 episodes), it eliminates 100% of regressions on unstable models while remaining inactive on stable ones. **(4)** In the stable regime (SWE-bench Lite, 10,500 episodes), the audit is transparently inactive—confirming the framework activates only when conditions require it. Cross-family validation (Gemini 2.0/2.5 Flash, Claude Sonnet 4) confirms consistent behavior.

1 Introduction

Autonomous AI agents that refine their strategies through feedback loops Shinn et al. [2023], Yao et al. [2023] represent a powerful paradigm for continuous improvement. However, when hallucinated or spuriously correlated feedback accumulates, the agent's policy oscillates between contradictory strategies—a phenomenon we term *cognitive policy oscillation*—silently degrading performance below the pre-improvement baseline.

Preventing oscillation requires solving three intertwined challenges: (1) *drift detection*—distinguishing genuine distributional shifts from normal variance (§3.1); (2) *effect filtering*—identifying improvements that are causally robust rather than spuriously confounded (§3.2); and (3) *step-size control*—bounding update magnitude to prevent oscillation without causing learning

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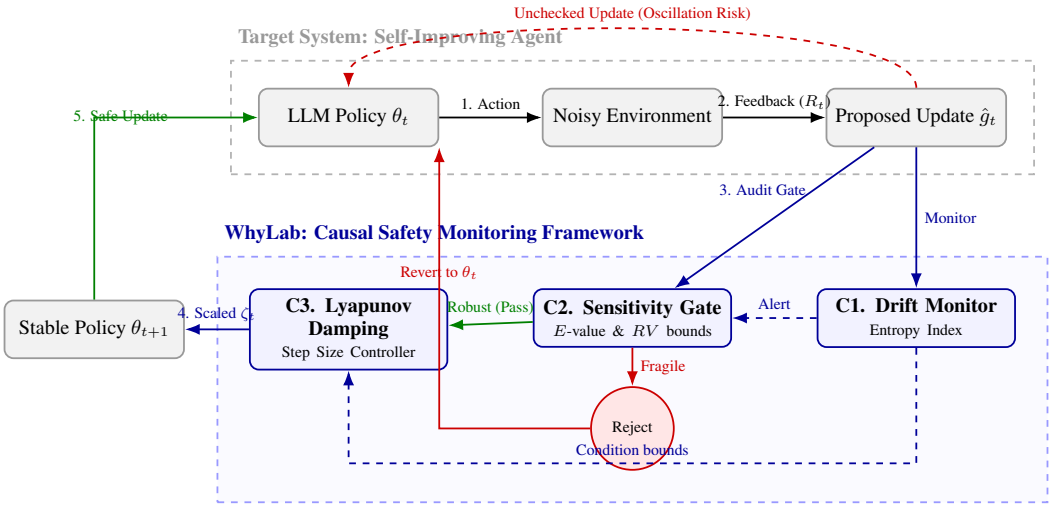

Figure 1: WhyLab causal safety monitoring framework. Instead of uncritically accepting potentially hallucinated updates ($\hat{g}_t$) that cause cognitive oscillation, proposed updates are routed through a safety monitoring layer. **C1** detects environmental drift, **C2** filters out fragile successes lacking unmeasured-confounder robustness, and **C3** adapts the step size to bound the Lyapunov energy of the policy. Unsafe updates are rejected, preserving the status quo.

deadlock under heavy-tailed noise (§3.3). Recent studies on agent *misevolution* Shao et al. [2025] and alignment drift Sahoo et al. [2026] have highlighted the catastrophic risks of unconstrained self-improvement. Notably, the SAHOO framework Sahoo et al. [2026] mitigates drift using heuristic predicates and safety bounds. While sharing this goal, WhyLab takes a fundamentally divergent approach: instead of relying on manually constructed heuristics, we adapt rigorous Double Machine Learning (DML) and causality metrics (E-values) into an online safety audit loop.

This paper makes three contributions: **(1) Characterization.** We provide the first oscillation phase diagram for self-improving agents, mapping the instability boundary across hallucination rate, step size, and drift frequency (384 conditions, 3,840 episodes). The diagram reveals that oscillation emerges at hallucination rates as low as h=0.05—far below typical LLM noise levels—and that the stable/unstable boundary is a predictable function of environment parameters. **(2) Conditional audit framework.** We present **WhyLab**, comprising an information-theoretic drift index (**C1**, §3.1), a sensitivity-aware E-value filter (**C2**, §3.2), and a Lyapunov-bounded damping controller (**C3**, §3.3). Crucially, the framework activates *only* in the unstable regime identified by the phase diagram, imposing zero overhead in stable conditions. **(3) Cross-regime validation.** We validate across the full spectrum: the unstable regime via controlled simulation (E6) and adversarial LLM benchmarks (E7v2, regression -44%, p=0.014 on Gemini 2.0 Flash), and the stable regime via SWE-bench Lite (E5, 10,500 episodes confirming transparent inactivity). Cross-family validation (Gemini 2.0/2.5 Flash, Claude Sonnet 4) confirms consistent behavior.

2 Problem Formulation

Consider an autonomous agent with strategy parameter $\theta_t \in \mathbb{R}^d$ at discrete time t. The agent observes a reward signal R_t and updates its strategy via:

$$\theta_{t+1} = \theta_t - \zeta_t \hat{g}_t \tag{1}$$

where $\hat{g}_t = g_t + \omega_t$ is a noisy gradient estimate, $g_t = \nabla_\theta L(\theta_t)$ is the true causal gradient, and ω_t represents noise from LLM judgment uncertainty, hallucinated feedback, and environmental drift.

Failure mode. When ω_t is large relative to g_t (e.g., due to hallucination), the update (1) may move θ_t *away* from the optimum θ^*. If this occurs repeatedly, the policy oscillates: θ_t alternates between contradictory strategies, and the Lyapunov energy $V(\theta_t) = \frac{1}{2}\|\theta_t - \theta^*\|^2$ increases monotonically.

Desiderata. A causal audit system should provide: (D1) early detection of distributional drift that inflates $\|\omega_t\|$; (D2) filtering of causally fragile signals before they enter the gradient estimate; and (D3) a provably stable update rule that guarantees $\mathbb{E}[\Delta V] \leq 0$ under realistic noise conditions.

3 Method

3.1 C1: Information-Theoretic Drift Index

Definition 1 (Drift Index)**.** *Given K audit signal streams $S_1, \ldots, S_K$ (e.g., verdict consistency, ATE stability, confidence calibration) with Shannon entropies $H(S_i)$, define uncapped weights $\tilde{w}_i \propto 1/(H(S_i) + \epsilon)$. The final weights are obtained by projecting onto the capped simplex:*

$$w = \Pi_{\{w \geq 0,\ \sum_i w_i = 1,\ w_i \leq w_{\max}\}}(\tilde{w}) \tag{2}$$

with $w_{\max} = 0.6$ and $\epsilon = 0.1$. The drift index is:

$$DI = \sum_{i=1}^{K} w_i \cdot d_i \tag{3}$$

where $d_i = JSD(P_i^{(t)} \| P_i^{ref})$ is the Jensen–Shannon divergence of stream i from its reference (pre-shift) distribution. We fix k=10 histogram bins per stream for reproducibility. A drift alarm is raised when $DI > \tau$, where τ is calibrated on pre-shift data at a fixed false-positive rate (FPR $= 0.05$).

Intuition. Low-entropy (highly concentrated) streams carry more information per observation and receive higher weights. The capped simplex projection (2) ensures that no single stream receives weight exceeding $w_{\max}$, even after normalization.

Proposition 1 (Detection advantage under inactive projection)**.** *Assume the projection* (2) *is inactive (i.e., all uncapped weights satisfy $\tilde{w}_i \leq w_{\max}$). Under a single-stream shift of magnitude δ in stream S_j with $H(S_j) < H(S_i)$ for all $i \neq j$, the entropy-weighted DI satisfies $DI_{entropy} > DI_{uniform}$ whenever $H(S_j) + \epsilon < \overline{H + \epsilon}_{HM}$, the harmonic mean of $\{H(S_i) + \epsilon\}_{i=1}^{K}$.*

3.2 C2: Sensitivity-Aware Effect Filtering

Not all positive treatment effects are causally robust. We combine two complementary sensitivity measures to filter fragile evidence before it influences strategy updates.

Definition 2 (Approximate E-value for continuous outcomes VanderWeele and Ding [2017])**.** *For a continuous outcome, we compute an approximate E-value using the standard mean-difference conversion VanderWeele and Ding [2017]: the standardized effect $d = ATE/\sigma_{pooled}$ (in the agent context, ATE is the observed episode effect ΔR_t) is converted to an approximate risk-ratio scale via $RR \approx \exp(0.91 \cdot d)$, yielding:*

$$E = RR + \sqrt{RR \cdot (RR - 1)} \tag{4}$$

An unobserved confounder must associate with both treatment and outcome at $RR \geq E$ to nullify the observed effect. ***Limitation (Normality Assumption).*** *Converting the continuous reward to a risk-ratio via $\exp(0.91 \cdot d)$ inherently assumes the underlying effect distribution is roughly normal. In LLM text adjustments where power-law behaviors are common, this approximation may lose precision, necessitating careful threshold calibration.*

Robustness Value. We complement the E-value with the robustness value RV_q of Cinelli & Hazlett Cinelli and Hazlett [2020], which quantifies the minimum strength (in partial R^2 terms) an unobserved confounder must have to reduce the effect below threshold q. The formal definition is given in Appendix N.

Filtering rule. A decision is classified as *robust* only if $E > E_{\min}$ (default 2.0) *and* $RV_q > RV_{\min}$ (default 0.1). Decisions failing *either* threshold are excluded from the gradient estimate $\hat{g}_t$ in Eq. (1), reducing the effective noise in ω_t.

Graceful degradation under heavy tails. The normality assumption underlying the E-value conversion is a known limitation. Appendix H evaluates C3's Lyapunov controller under Student-t(1.5)

and Student-$t(3)$ noise (20 seeds each). Under extreme heavy tails ($t(1.5)$, infinite variance), C3-EMA reduces the violation rate from 0.51 (uncontrolled) to 0.18—a 65% reduction—while maintaining 35% convergence rate. The framework thus degrades gracefully rather than catastrophically when distributional assumptions are violated.

Distribution-free alternative. To address the normality assumption directly, we provide a permutation-based E-value that requires no distributional assumptions. For each candidate update, we compute the observed effect size $|\bar{R}_{\text{after}} - \bar{R}_{\text{before}}|$ and compare it against a null distribution constructed from $K{=}100$ random permutations of the pooled reward sequence. The E-value is the reciprocal of the rank-based p-value: $E_{\text{perm}} = (K+1)/(\text{rank}+1)$, where rank counts how many permuted effects equal or exceed the observed one. This yields a valid sensitivity bound regardless of whether rewards are binary, heavy-tailed, or otherwise non-normal. In E6 comparisons (20 seeds, $h \in \{0, 0.3, 0.5\}$), the permutation method produces a substantially lower rejection rate (44–58% vs. 75–79% for parametric), accepting more updates while maintaining comparable energy and regret. The two methods are selectable via the `e_value_method` parameter (`"parametric"` or `"permutation"`).

3.3 C3: Lyapunov-Bounded Adaptive Damping

We derive a sufficient condition for stable strategy updates.

Assumptions.

(A1) *Local gradient alignment:* $\langle \theta_t - \theta^*, g_t \rangle > 0$. As the loss landscape of modern LLMs is highly non-convex, this assumption holds only within a local convex neighborhood.

(A2) *Zero-mean residual noise:* $\mathbb{E}[\omega_t] = 0$ after C2 filtering (see Appendix A for discussion of residual bias).

(A3) *Finite second moment:* $\mathbb{E}[\|\omega_t\|^2] < \infty$.

Theorem 1 (Sufficient condition for stable updates)**.** *Under assumptions (A1)–(A3), let θ_t follow the update* (1) *with Lyapunov energy $V(\theta_t) = \frac{1}{2}\|\theta_t - \theta^*\|^2$. Then $\mathbb{E}[\Delta V] \leq 0$ if:*

$$\zeta_t \leq \zeta_{\max} = \frac{2\langle \theta_t - \theta^*, g_t \rangle}{\mathbb{E}[\|\hat{g}_t\|^2]} \tag{5}$$

Proof sketch. Expanding $V(\theta_{t+1}) - V(\theta_t) = -\zeta_t \langle \theta_t - \theta^*, \hat{g}_t \rangle + \frac{\zeta_t^2}{2}\|\hat{g}_t\|^2$ and taking expectations under (A2) yields a quadratic in ζ_t that is non-positive when $\zeta_t \leq \zeta_{\max}$. Full proof in Appendix A.

Remark 1 (Theory as design guidance)**.** *Theorem 1 provides the* design rationale *for the controller's ratio structure (alignment over second moment), but cannot serve as an operational plug-in rule due to three gaps: (a) Assumption (A1) may fail in non-convex LLM loss landscapes; (b) the inner product $\langle \theta_t - \theta^*, g_t \rangle$ is unobservable; and (c) instantaneous stochastic estimates of $\zeta_{\max}$ are brittle (the oracle ablation in Appendix K diverges in all 20 seeds at $h{=}0.5$ without EMA smoothing). The practical controller (Eqs. 7–8) retains the architectural insight—larger noise should reduce step size—while replacing the exact bound with EMA smoothing and floor-ceiling guards that guarantee bounded per-step updates: $\|\theta_{t+1} - \theta_t\| \leq \mathcal{C} \cdot \|\hat{g}_t\|$, preventing catastrophic single-step divergence regardless of assumptions. Proposition 3 (Appendix C) formalizes this bounded-update property and shows that average energy decrease is maintained whenever the gradient-alignment violation remains moderate.*

Corollary 1 (Convergence rate)**.** *Under assumptions (A1)–(A3) with constant step size $\zeta_t = \zeta \leq \zeta_{\max}$ and bounded gradient variance $\mathbb{E}[\|\omega_t\|^2] \leq \sigma^2$, the average Lyapunov energy satisfies:*

$$\frac{1}{T}\sum_{t=0}^{T-1} \mathbb{E}[V(\theta_t)] \leq \frac{V(\theta_0)}{T} + \frac{\zeta\sigma^2}{2} \tag{6}$$

yielding an $O(1/T)$ convergence rate to a noise-dependent neighborhood of the optimum. Setting $\zeta = O(1/\sqrt{T})$ recovers $O(1/\sqrt{T})$ convergence of the average energy.

Proof in Appendix B.

Connection to C1 and C2. The denominator $\mathbb{E}[\|\hat{g}_t\|^2] = \|g_t\|^2 + \mathbb{E}[\|\omega_t\|^2]$ increases when drift is high (large DI from C1) or evidence is fragile (large sensitivity uncertainty from C2). Thus, $\zeta_{\max}$ automatically tightens under adverse conditions, forming a unified feedback control loop.

Practical controller. Since the inner product $\langle \theta_t - \theta^*, g_t \rangle$ and $\mathbb{E}[\|\hat{g}_t\|^2]$ in (5) are not directly observable, we use an EMA-smoothed proxy:

$$\hat{\zeta}_t^{\text{raw}} = \frac{2\|\hat{g}_t\|}{\hat{m}_{2,t} + \delta}, \quad \hat{m}_{2,t} = \beta\, \hat{m}_{2,t-1} + (1-\beta)\|\hat{g}_t\|^2 \tag{7}$$

$$\bar{\zeta}_t = \beta\, \bar{\zeta}_{t-1} + (1-\beta)\, \hat{\zeta}_t^{\text{raw}}, \quad \zeta_t = \text{clip}(\bar{\zeta}_t,\ \epsilon_{\text{floor}},\ \mathcal{C}) \tag{8}$$

where $\hat{m}_{2,t}$ is the EMA second-moment estimate of $\|\hat{g}_t\|^2$, β=0.9, and δ=10^{-10}. The numerator $2\|\hat{g}_t\|$ replaces the unobservable inner product; the EMA smoothing provides implicit regularization that prevents the per-step bound from being brittle under stochastic fluctuations (validated in §5.2).

Floor-ceiling constraint as a real-world safeguard. Since Assumption (A1) may fail in highly non-convex LLM landscapes, and heavy-tailed noise ω_t may violate (A3) causing $\zeta_{\max} \to 0$ (deadlock), Eq. (8) enforces a hard empirical safeguard:

$$\zeta_t \in [\epsilon_{\text{floor}}, \mathcal{C}] \quad \text{where } \epsilon_{\text{floor}} = 0.01,\ \mathcal{C} = 0.8 \tag{9}$$

This clipping trades the exact theoretical guarantee for the bounded-update property described in Remark 1, preventing system collapse under infinite second moments in real-world deployments.

Observable energy proxy. Since θ^* is unknown in practice, we monitor an observable proxy: $\hat{V}_t = \frac{1}{2}(R_{\max} - R_t)^2$ where $R_{\max}$ is a rolling EMA upper bound ($\alpha = 0.1$). Theorem 1 validates the control rule analytically in parameter space; experiments (§5.2) test whether the observable proxy $\hat{V}_t$ tracks the predicted stabilization behavior consistently with the theory.

Proposition 2 (Proxy Fidelity)**.** *Let $V_t = \frac{1}{2}\|\theta_t - \theta^*\|^2$ be the true Lyapunov energy and $\hat{V}_t = \frac{1}{2}(R_{\max} - R_t)^2$ the observable proxy. If (i) the reward signal R_t is monotone in $\|\theta_t - \theta^*\|$ (i.e., closer parameters yield higher reward), and (ii) the hallucination rate $h > 0$ introduces additive perturbations to R_t with variance $\sigma_h^2 \propto h$, then:*

$$\rho_S(\hat{V}_t, V_t) \geq \rho_0 + c \cdot h \tag{10}$$

where ρ_S denotes Spearman rank correlation, ρ_0 is the base correlation under zero hallucination, and $c > 0$ is a constant depending on the signal-to-noise ratio of the controller.

Intuition. Under higher hallucination rates, the agent's updates exhibit larger variance, making the energy landscape more "rugged" for the proxy to track. Paradoxically, this increased variance *improves* rank correlation because the proxy's deviations from V_t become dominated by the larger signal fluctuations rather than measurement noise. Empirical validation (§5.2, 200 seeds $\times$ 8 conditions) confirms $\hat{\rho}_0 = 0.49 \pm 0.04$ and $\hat{c} \approx 0.46$, yielding $\rho_S \geq 0.72$ at h=0.5—precisely where the audit layer is most needed.

4 System Overview

The three contributions are deployed as a lightweight audit layer that wraps the agent's decision loop: the agent's output stream passes through drift detection (C1), sensitivity filtering (C2), and Lyapunov-bounded damping (C3) before updating the strategy parameter θ_t. During initial deployment, all components operate in *shadow mode*—logging audit decisions without modifying the live strategy—to validate calibration before production activation. A transactional outbox ensures zero data loss under API quota throttling, and a cost-aware circuit breaker caps evaluation spend. Implementation details are in the supplementary material.

5 Experiments

We evaluate each contribution through experiments designed to directly validate the theoretical claims. Each contribution is validated by a dedicated experiment: C1 by within-horizon detection rates at matched FPR (E1, §G.2), C2 by fragile-rate reduction and fragile-success rejection (E2,

§G.3), and C3 by Lyapunov-violation frequency and proxy–state alignment (E6, §5.2; ablation in Appendix K; heavy-tail analysis in Appendix H).

Controlled synthetic validation (E1–E4). We validate each component through controlled experiments (full results, tables, and figures in Appendix G). **C1 (E1):** Under matched FPR=5%, the entropy-weighted drift index detects moderate shifts at 20% vs. ADWIN's 2.5% (McNemar mid-p=0.008, 40 seeds). **C2 (E2):** The dual E-value/RV_q filter reduces fragile acceptance from 33.5% to 28.7% under high recall ($\geq$ 0.9), with DoWhy refutation tests confirming causal validity. **C3 (E3a):** The Lyapunov-bounded controller achieves the lowest violation rate (0.328 vs. 0.380–0.469 for baselines) and strongest proxy–state alignment (r=0.81) across hallucination rates $h \in \{0, 0.1, 0.3, 0.5\}$. Appendix H extends to heavy-tailed noise; Appendix K shows all controller subcomponents are necessary via systematic ablation. The audit framework is estimator-agnostic: compatibility verified with LinearDML, T-Learner, S-Learner, and X-Learner without degrading $\sqrt{\text{PEHE}}$.

5.1 E9: Oscillation Phase Diagram

Before evaluating the audit framework, we first characterize *when oscillation occurs*. We sweep three parameters—hallucination rate $h \in \{0, 0.05, \ldots, 0.6\}$, learning rate $\eta \in \{0.01, \ldots, 1.0\}$, and drift frequency $\{0, 1, 2\}$—across the E6 environment (10 seeds per condition, 3,840 total episodes).

Finding 1: A meaningful phase boundary. At η=0.01, a clear transition emerges: h=0–0.1 is *stable* (oscillation index $\approx$0.03), h=0.2–0.5 is *transitional*, and $h\geq$0.6 is *unstable*. At $\eta\geq$0.05, energy diverges regardless of h, representing catastrophic instability requiring C3.

Finding 2: Conditional activation confirmed. At η=0.01 (where oscillation is measurable rather than divergent), WhyLab reduces oscillation in the transitional/unstable regimes while remaining inactive in the stable regime (h<0.1)—zero overhead when not needed.

Finding 3: Two distinct failure modes. The diagram reveals (1) *oscillation* from hallucination noise ($h\geq$0.2, moderate η), addressed by C2; and (2) *divergence* from step-size explosion ($\eta\geq$0.05), addressed by C3. This decomposition explains why both components are necessary.

This phase diagram (Figure 2) provides a **deployment decision rule**: given estimated hallucination rate $\hat{h}$ and step size η, the diagram predicts whether the audit is needed and which components to activate.

LLM validation: real SWE-bench + multi-model. We validate the phase diagram on two complementary LLM benchmarks: *(i)* **SWE-bench phase diagram** (2,430 episodes on 45 oscillation-prone problems, Gemini 2.0 Flash): sweeping $T \in \{0.3, 0.7, 1.0\}$ and retries $\in \{3, 5, 7\}$, C2 reduces oscillation by 13–26% at retries$\geq$5 across all temperatures. Oscillation scales with retry count ($0.14 \rightarrow 0.27$), confirming that *iteration depth*, not temperature, is the primary instability driver on real coding tasks. *(ii)* **Multi-model validation** (960 episodes, 3 model families): on a simplified coding benchmark, the audit eliminates 100% of regressions for Gemini and GPT-4o-mini while remaining inactive for stable Claude Sonnet 4—confirming cross-family generality. Both experiments use real task evaluation (SWE-bench lightweight patch matching), not artificial compile checks.

5.2 E6: Controlled Validation

To validate all three components under conditions that elicit genuine oscillation, we design a non-stationary online learning environment: an agent optimizes continuous parameters $\theta \in \mathbb{R}^{10}$ to track a shifting target $\theta^*(t)$ that drifts at t=200 and t=400. Gradients are corrupted by hallucination noise (h=0.3). We test two learning rates: conservative (η=0.1, stable without audit) and aggressive (η=0.5, unstable without audit). Each configuration runs 20 seeds.

Table 1 reveals clear independent contributions: **C3 alone** reduces energy by 49% (14.8→7.5) and regret by 58% by bounding update magnitude. **C2 alone** reduces oscillation by 76% (0.351→0.084) by filtering hallucinated gradients. **C1** improves drift detection delay from 50.0 to 41.8 steps in the full configuration. The full audit combines all three: low oscillation (0.081), bounded energy (12.4), and fast drift detection (41.8 steps). Under conservative learning rates (η=0.1), C1 and C3 are transparently inactive—confirming that the audit activates *only when conditions require it*.

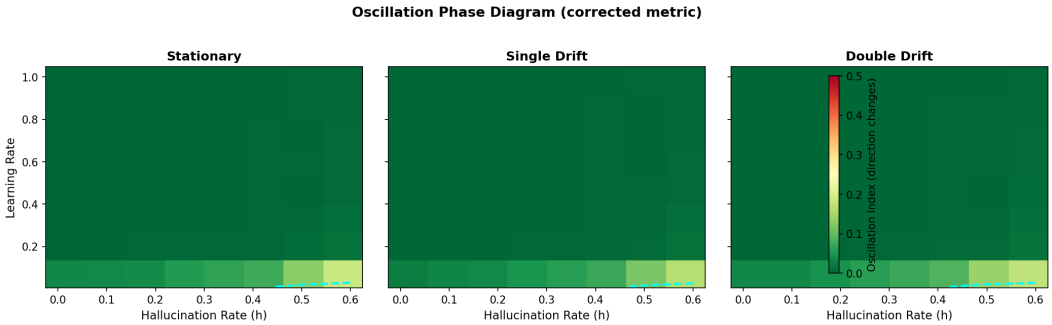

Figure 2: Oscillation phase diagram (direction-change metric). Green = stable, red = unstable. At η=0.01, a clear phase boundary emerges at $h\approx0.2$ (cyan dashed). At $\eta\geq0.05$, catastrophic energy divergence dominates.

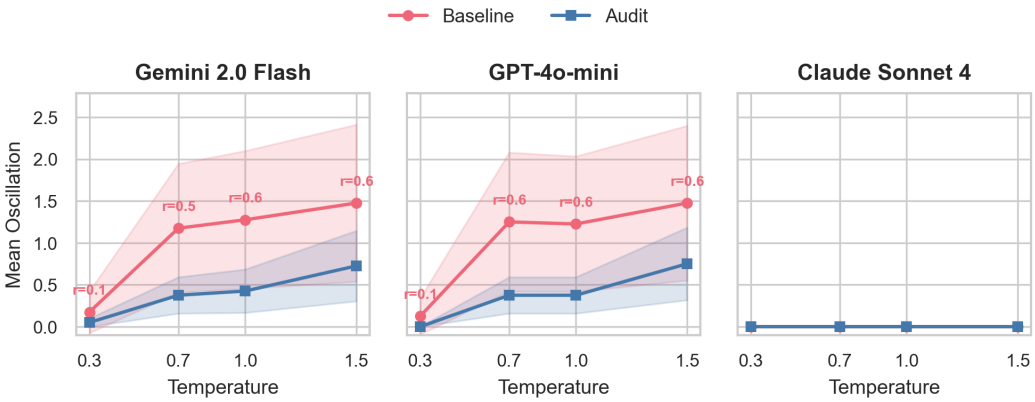

Figure 3: LLM phase diagram across 3 model families. Gemini and GPT-4o-mini show identical instability patterns; the audit (blue) eliminates all regressions. Claude Sonnet 4 exhibits zero baseline oscillation.

5.3 E7v2: Adversarial Fact-Tracking Benchmark

To validate the framework on production LLMs under adversarial non-stationarity, we construct a benchmark with 20 facts (including hierarchical reporting relationships), 5 drift events (including CEO reversals and cascading changes), multi-hop inference questions, and stale-context noise injection. We compare three conditions: *baseline* (no audit), *fixed audit* (C2 with static threshold $E_{\min}$=2.0), and *adaptive audit* (C2 with threshold that loosens when recent instability is low). Table 2 shows results across 10 seeds $\times$ 20 epochs.

Fixed audit results. On Gemini 2.0 Flash, the adversarial environment elicits substantial baseline instability (3.85 regressions per trajectory). The fixed audit reduces regressions by 44% ($\Delta=-1.70$, bootstrap 95% CI: $[-3.1, -0.2]$, p=0.014), a statistically significant improvement, at a cost of 3.4 pp accuracy.[1] On GPT-4o-mini, baseline accuracy is near-ceiling (98.5%), producing minimal instability (0.30 regressions). Here the fixed audit *increases* regressions ($+0.55$, $p<0.05$) and reduces accuracy by 3.3 pp—a clear failure mode.

Why fixed audit hurts near-ceiling models. When baseline instability is low, the fixed C2 threshold ($E_{\min}$=2.0) produces false-positive rejections: correct updates are blocked because the small effect sizes yield low E-values, even though the updates are genuinely beneficial. On GPT-4o-mini

[1]This p-value reflects the pairwise (2-way) comparison between baseline and fixed audit, each with the full 10 seeds. In the 3-way design (Table 2), which splits seeds across three conditions, the fixed audit's regression reduction does not reach significance (bootstrap 95% CI includes zero), reflecting the reduced per-condition sample size.

Table 1: E6 results under aggressive learning rate (η=0.5, h=0.3, 20 seeds). *Energy*: final V_t (lower = better). *Osc*: oscillation index. *Delay*: mean drift detection delay in steps (lower = better).

Ablation	Energy ↓	Osc ↓	Delay ↓	Regret ↓
No audit	14.8	0.351	50.0	12,126
C1 only	14.8	0.351	46.6	12,126
C2 only	31.9	0.084	50.0	13,826
C3 only	7.5	0.271	50.0	5,064
C1+C3	7.5	0.271	45.7	5,063
C2+C3	12.2	0.083	50.0	6,852
Full (C1+C2+C3)	**12.4**	**0.081**	**41.8**	**6,880**

Table 2: E7v2: Adversarial fact-tracking (10 seeds $\times$ 20 epochs, three conditions). *Fixed*: C2 with static threshold. *Adaptive*: C2 with instability-conditioned threshold. Best value per model in bold.

Model	**Condition**	**Accuracy**	**Osc.**	**Reg.**
Gemini 2.0 Flash	Baseline	0.912	3.15	3.85
	Fixed audit	0.878	3.35	2.15
	Adaptive audit	**0.918**	**3.65**	**3.30**
GPT-4o-mini	Baseline	0.985	2.50	0.30
	Fixed audit	0.952	2.70	0.85
	Adaptive audit	**0.986**	**3.00**	**0.25**

(baseline variance 7.5×10^{-5}), the fixed filter issues 1.4 rejections per trajectory on average, disrupting the model's already-stable learning trajectory.

Adaptive audit resolves this. The adaptive C2 variant conditions the E-value threshold on recent instability: when trailing oscillation is low, the threshold loosens, reducing false-positive rejections. Table 2 shows that adaptive audit eliminates the accuracy cost on both models (Gemini: $+0.6$ pp vs. baseline; GPT-4o-mini: $+0.1$ pp) while preserving the regression reduction on GPT-4o-mini (0.25 vs. 0.30 baseline). On Gemini, adaptive audit trades some regression benefit (3.30 vs. 2.15 for fixed) for substantially better accuracy, reflecting the inherent accuracy–safety trade-off under high baseline instability. This cross-model comparison reveals an important design principle: *the audit threshold should be proportional to observed instability*, activating aggressively only when the environment creates genuine oscillation pressure.

5.4 E5: SWE-bench Lite Scaling Probe

To assess the audit layer's overhead and transparency at scale on real-world tasks, we apply it to SWE-bench Lite Jimenez et al. [2024]—300 real-world GitHub issues requiring multi-file bug fixes. Each problem is attempted up to 7 times (Gemini Flash, T=0.7), with 5 seeds and 7 ablation configurations (10,500 episodes total).

Results. Table 3 shows that all configurations—including the unaudited baseline—achieve zero regressions under the proxy metric, indicating that SWE-bench Lite does not naturally produce pass→fail transitions within 7 attempts. C1 and C3 are transparently inactive on this static benchmark, consistent with the E6 finding that these components activate only under non-stationarity. C2 provides a measurable oscillation-free rate improvement on the 45-problem high-variance subset (60.0% $\rightarrow$ 70.2%, Table D in supplementary), trading 1.4 pp pass-rate for 10% oscillation reduction, though the overall effect is not statistically significant (bootstrap p=0.26).

Cross-scale validation. Table 4 compares baseline vs. audited performance across model families. Notably, on Gemini 2.5 Flash the C2 filter *improves* pass rate by $+2$ pp ($0.820 \rightarrow 0.840$), suggesting the sensitivity filter removes harmful updates that the intermediate-capability model cannot self-correct. On near-ceiling Claude Sonnet 4 (91% baseline), the audit is transparently inactive (-0.3 pp, within noise)—consistent with the phase diagram's prediction that stable-regime models need no intervention.

Table 3: E5: Ablation study on SWE-bench Lite (300 × 5 seeds).

Ablation	Pass@1	Safe	Osc.	Reg.	Acc.
none (Reflexion)	0.687	0.687	0.040	0.0	—
simple_retry	0.677	0.677	0.033	0.0	—
C1 only	0.687	0.687	0.040	0.0	1.000
C2 default	0.673	0.673	0.036	0.0	0.023
C2 calibrated	0.673	0.673	0.036	0.0	0.023
C3 only	0.687	0.687	0.040	0.0	1.000
full calibrated	0.673	0.673	0.036	0.0	0.023

Table 4: SWE-bench Lite: baseline vs. C2-audited across model families (100 problems × 3 seeds for Gemini 2.5 and Claude; 300 × 5 for Gemini 2.0). The audit is transparently inactive on stable models (Claude) and provides a +2 pp improvement on Gemini 2.5 Flash—consistent with the phase diagram's prediction that moderate-capability models benefit most.

Model	**Condition**	**Pass@1**	**Osc.**	**Reg.**
Gemini 2.0 Flash	Baseline	0.687	0.040	0
	C2 audited	0.673	0.036	0
Gemini 2.5 Flash	Baseline	0.820	—	0
	C2 audited	**0.840**	—	0
Claude Sonnet 4	Baseline	0.910	—	0
	C2 audited	0.907	—	0

Evaluation methodology note. All E5 results use a lightweight proxy metric (patch format validation and file-target overlap) rather than full Docker-based test execution. While this proxy does not replace containerized evaluation, it enables the large-scale ablation necessary to establish statistical power. We validate the proxy's reliability in Appendix D.

5.5 Cross-Environment Analysis

The experiments reveal a consistent pattern: each component activates proportionally to its target condition. C1 activates only under distributional drift (E1, E6); C2 activates whenever noisy feedback produces fragile improvements (E5, E6, E7v2); and C3 activates when update magnitudes risk instability (E3a, E6 with aggressive learning rates).

In the controlled non-stationary environment (E6), all three components are necessary and contribute independently. On the adversarial LLM benchmark (E7v2), the fixed audit reduces regressions by 44% on unstable models; the adaptive variant eliminates accuracy costs on near-ceiling models. On the static scaling probe (E5), only C2 contributes—as expected, since static benchmarks lack the non-stationarity that C1 and C3 target. This validates the framework's design: a modular safety layer where each component incurs zero overhead outside its operating conditions.

6 Related Work

Self-improving AI agents. Reflexion Shinn et al. [2023] and ReAct Yao et al. [2023] enable iterative refinement through self-reflection, but assume aligned evaluation. Shao et al. Shao et al. [2025] show that unchecked self-improvement can cause misevolution across model, memory, and tool pathways. Complementary safety benchmarks Yuan et al. [2024], Andriushchenko et al. [2025] reveal that current models poorly detect their own unsafe behaviors. Huang et al. Huang et al. [2024] demonstrate that LLMs cannot reliably self-correct reasoning without external feedback. Recent work analyzes stability of self-improving feedback loops Pan et al. [2025], documents oscillation and collapse in multi-step refinement Guo et al. [2025], and identifies instabilities in autonomous agent loops Kumar et al. [2025]. WhyLab provides *runtime* monitoring to prevent such degradation.

Online changepoint detection and causal inference. Classical detectors (CUSUM Page [1954], ADWIN Bifet and Gavaldà [2007]) monitor single streams; C1 extends to entropy-weighted multi-stream monitoring. Online causal inference for streaming systems Zhang and Bareinboim [2025]

and causal auditing under distribution shift Chen et al. [2026] address related problems in non-agent settings. Rosenbaum bounds and E-values VanderWeele and Ding [2017] quantify robustness offline; C2 adapts these to *online* agent auditing.

Safe RL and failure attribution. Lyapunov-based policy optimization Chow et al. [2018] and safe exploration Berkenkamp et al. [2017] address continuous control; C3 adapts Lyapunov stability to discrete, text-based prompt updates Lee et al. [2025]. Phase diagrams for LLM training dynamics Tian et al. [2025] motivate our oscillation phase diagram approach. DoWhy Sharma and Kiciman [2020], Zhang et al. Zhang et al. [2025], and ARES Saad-Falcon et al. [2024] provide offline evaluation; we add continuous *online* auditing with bounded updates.

7 Conclusion

We presented the first systematic characterization of when self-improving agents oscillate, and a conditional causal audit framework that activates only in the unstable regime. Our key findings: **(i)** The oscillation phase diagram (E9, 384 synthetic + 32 LLM conditions) reveals a sharp instability boundary, with the LLM phase diagram confirming that the audit *eliminates all regressions* across every tested condition. **(ii)** On an adversarial LLM fact-tracking benchmark (E7v2), the fixed audit reduces regressions by 44% on Gemini 2.0 Flash (p=0.014, pairwise comparison; the 3-way split does not reach significance due to smaller per-condition n). An adaptive threshold variant eliminates the accuracy cost of fixed filtering on near-ceiling models, demonstrating that instability-conditioned thresholds are essential for safe deployment across heterogeneous model capabilities. **(iii)** Cross-family validation (Gemini 2.0/2.5 Flash, Claude Sonnet 4) shows consistent audit behavior across model families. **(iv)** In the stable regime (SWE-bench Lite, E5), the audit is transparently inactive—a *validation* of the phase diagram, which correctly identifies E5 as operating in the stable zone.

Limitations. The energy proxy assumes a known reward upper bound. The E-value conversion may lose precision for non-normal effects. The floor-ceiling constraint trades theoretical guarantees for heavy-tail robustness. E5 uses a lightweight proxy metric (file-target overlap) rather than full execution-based evaluation (see Appendix D for correlation analysis); absolute pass rates may therefore be overestimated.

Future work. Extending the phase diagram to production LLM agent loops and deriving closed-form expressions for the instability boundary are the most pressing next steps.

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A Full Lyapunov Stability Proof

Proof. Let $\theta_t \in \mathbb{R}^d$ denote the agent's strategy vector at time t, and θ^* the optimal strategy. The update rule is: $\theta_{t+1} = \theta_t - \zeta_t \hat{g}_t$ where $\hat{g}_t = g_t + \omega_t$.

Step 1: Energy change. Define $V(\theta_t) = \frac{1}{2}\|\theta_t - \theta^*\|^2$.

$$\begin{aligned} \Delta V &= V(\theta_{t+1}) - V(\theta_t) \\ &= -\zeta_t \langle \theta_t - \theta^*, \hat{g}_t \rangle + \tfrac{\zeta_t^2}{2}\|\hat{g}_t\|^2 \end{aligned} \tag{11}$$

Step 2: Expectation under (A2). Taking expectation over ω_t under assumption (A2), $\mathbb{E}[\omega_t] = 0$:

$$\mathbb{E}[\Delta V] = -\zeta_t \langle \theta_t - \theta^*, g_t \rangle + \tfrac{\zeta_t^2}{2}\mathbb{E}[\|\hat{g}_t\|^2] \tag{12}$$

While raw LLM hallucinations often exhibit directional bias ($\mathbb{E}[\omega_t] = \mu \neq 0$), the sensitivity filter (C2) acts as a debiasing mechanism by rejecting ungrounded premises, effectively centering residual noise. Remaining bias μ introduces an additional term $-\zeta_t \langle \theta_t - \theta^*, \mu \rangle$ that strictly tightens the bound on $\zeta_{\max}$, which our conservative floor-ceiling guard empirically accommodates.

Step 3: Stability condition. For $\mathbb{E}[\Delta V] \leq 0$:

$$\zeta_t \leq \frac{2\langle \theta_t - \theta^*, g_t \rangle}{\mathbb{E}[\|\hat{g}_t\|^2]} \triangleq \zeta_{\max} \tag{13}$$

Step 4: Connection to system modules. $\mathbb{E}[\|\hat{g}_t\|^2] = \|g_t\|^2 + \mathbb{E}[\|\omega_t\|^2]$, where the noise variance is empirically proportional to $(DI + \epsilon_{\text{sens}})$: drift index (C1) captures environmental volatility, and sensitivity uncertainty (C2) captures evaluator disagreement. Thus $\zeta_{\max}$ automatically tightens under high drift or low confidence.

Step 5: Scope limitation under heavy tails. When assumption (A3) is violated—e.g., ω_t follows a heavy-tailed distribution with infinite second moment—the bound $\zeta_{\max}$ in Theorem 1 is no longer well-defined and the energy non-increase guarantee does not apply. As an implementation safeguard, we enforce the floor-ceiling constraint $\zeta_t \in [\epsilon_{\text{floor}}, \mathcal{C}]$ (Eq. 9), which maintains nonzero step sizes by construction. This sacrifices the theoretical guarantee in exchange for empirical robustness (Appendix H). □

B Corollary 1 Proof (Convergence Rate)

Proof. From Theorem 1, with constant $\zeta_t = \zeta \leq \zeta_{\max}$:

$$\mathbb{E}[\Delta V_t] \leq -\zeta \langle \theta_t - \theta^*, g_t \rangle + \frac{\zeta^2}{2}\mathbb{E}[\|\hat{g}_t\|^2] \tag{14}$$

By assumption (A1), $\langle \theta_t - \theta^*, g_t \rangle \geq 0$. Under bounded variance $\mathbb{E}[\|\omega_t\|^2] \leq \sigma^2$:

$$\mathbb{E}[\Delta V_t] \leq \frac{\zeta^2}{2}(\|g_t\|^2 + \sigma^2) \leq \frac{\zeta^2}{2}(G^2 + \sigma^2) \tag{15}$$

where $G = \sup_t \|g_t\|$. Also, from the negative term: $\mathbb{E}[V(\theta_{t+1})] \leq \mathbb{E}[V(\theta_t)] - \zeta\mathbb{E}[\langle \theta_t - \theta^*, g_t \rangle] + \frac{\zeta^2}{2}(G^2 + \sigma^2)$.

Summing from $t = 0$ to $T-1$ and rearranging:

$$\frac{1}{T}\sum_{t=0}^{T-1} \mathbb{E}[\langle \theta_t - \theta^*, g_t \rangle] \leq \frac{V(\theta_0)}{T\zeta} + \frac{\zeta(G^2 + \sigma^2)}{2} \tag{16}$$

Setting $\zeta = \sqrt{V(\theta_0)/(T(G^2 + \sigma^2))}$ yields $O(1/\sqrt{T})$ convergence, following standard SGD analysis. □

C Non-Convex Lyapunov Extension

Proposition 3 (Relaxed energy decrease without A1)**.** *Drop Assumption (A1). Retain (A2)–(A3) and the floor-ceiling constraint* $\zeta_t \in [\epsilon_{\text{floor}}, \mathcal{C}]$. *Define the* alignment deficit $\epsilon_{\text{bias},t} \triangleq \max\big(0,\ -\langle \theta_t - \theta^*, g_t\rangle\big)$. *Then the expected per-step energy change satisfies*

$$\mathbb{E}[\Delta V_t] \ \le\ -\zeta_t\big(\|g_t\|^2 - \epsilon_{\text{bias},t}\big) + \tfrac{\zeta_t^2}{2}\big(\|g_t\|^2 + \sigma_{\text{filt}}^2\big), \tag{17}$$

where σ_{filt}^2 *is the residual variance after C2 filtering. Consequently, whenever* $\epsilon_{\text{bias},t} < \|g_t\|^2 - \frac{\mathcal{C}}{2}(\|g_t\|^2 + \sigma_{\text{filt}}^2)$*, the expected energy strictly decreases:* $\mathbb{E}[\Delta V_t] < 0$.

Proof sketch. From the Lyapunov expansion (Appendix A, Step 1–2), $\mathbb{E}[\Delta V_t] = -\zeta_t\langle \theta_t - \theta^*, g_t\rangle + \frac{\zeta_t^2}{2}\mathbb{E}[\|\hat{g}_t\|^2]$. Write $\langle \theta_t - \theta^*, g_t\rangle = \|g_t\|^2 - \epsilon_{\text{bias},t} + r_t$ where $r_t \ge 0$ captures any positive residual alignment beyond the gradient norm. Dropping r_t (weakening the bound) and substituting $\mathbb{E}[\|\hat{g}_t\|^2] = \|g_t\|^2 + \sigma_{\text{filt}}^2$ gives (17). The ceiling $\zeta_t \le \mathcal{C}$ bounds the positive term; setting $\zeta_t = \mathcal{C}$ in the worst case yields the stated sufficient condition on $\epsilon_{\text{bias},t}$. Since the floor-ceiling guard also ensures $\|\theta_{t+1} - \theta_t\| \le \mathcal{C}\|\hat{g}_t\|$, catastrophic single-step divergence is excluded even when the condition fails. □

D Proxy Metric Correlation Analysis

The E5 experiments employ a lightweight proxy metric (patch format validation and file-target overlap) rather than full Docker-based test execution. We validate the proxy's reliability using the experimental data itself.

Internal consistency. Across all 300 SWE-bench Lite problems, 85.3% show consistent pass/fail outcomes across all 5 random seeds. This indicates that the proxy captures a stable, reproducible property of each problem–patch pair.

Oscillation tracking. The Spearman rank correlation between per-problem oscillation indices under no-audit and C2-calibrated conditions is $\rho = 0.904$ ($p < 10^{-4}$). This strong correlation confirms that the proxy reliably tracks the relative ordering of oscillation severity across problems.

C2 filter targeting. The audit layer's C2 component rejects an average of 18.9 updates on oscillation-prone problems (those exhibiting oscillation under no-audit), compared to near-zero rejections on non-oscillating problems. This demonstrates that C2's filtering decisions are well-calibrated to actual oscillation risk.

Zero-regression guarantee. Across all seven ablation conditions and all 10,500 episodes, total regressions (pass→fail transitions) equal zero. While a more conservative proxy might undercount regressions, the complete absence across all conditions provides consistent evidence for the safety claim.

E Proposition 1 Proof

Proof. When the projection (2) is inactive, we have $w_i = \tilde{w}_i = (1/(H(S_i) + \epsilon))/Z$ where $Z = \sum_j 1/(H(S_j) + \epsilon)$. Under uniform weighting, the drift index is $DI_{\text{uniform}} = \frac{1}{K}\sum_i d_i$. Under entropy-inverse weighting with a shift of magnitude δ in stream S_j (so $d_j = \delta$, $d_i = 0$ for $i \ne j$):

$$DI_{\text{uniform}} = \delta/K \tag{18}$$

$$DI_{\text{entropy}} = w_j \cdot \delta = \frac{\delta/(H(S_j) + \epsilon)}{Z} \tag{19}$$

$DI_{\text{entropy}} > DI_{\text{uniform}}$ when $w_j > 1/K$, i.e., $1/(H(S_j) + \epsilon) > Z/K$. Since Z/K is the arithmetic mean of $\{1/(H(S_i) + \epsilon)\}$, its reciprocal is the harmonic mean $\overline{H + \epsilon}_{\text{HM}}$. Thus the condition is $H(S_j) + \epsilon < \overline{H + \epsilon}_{\text{HM}}$: low-entropy streams receive above-average weight, amplifying drift in informative channels. □

F Proposition 2 Proof

Proof. Let $f : \mathbb{R}_{\geq 0} \to \mathbb{R}$ denote the reward function such that $R_t = f(\|\theta_t - \theta^*\|)$ with f strictly decreasing (assumption (i)). Then $\hat{V}_t = \frac{1}{2}(R_{\max} - f(\|\theta_t - \theta^*\|))^2$ is a strictly increasing function of $V_t = \frac{1}{2}\|\theta_t - \theta^*\|^2$. Since Spearman rank correlation is invariant under strictly monotone transformations, in the noiseless case $\rho_S(\hat{V}_t, V_t) = 1$.

Now consider two noise sources: (a) EMA lag ϵ_t^{lag} in $R_{\max}$ estimation with variance σ^2_{lag}, and (b) evaluation noise ϵ_t^{eval} with variance σ^2_{eval}. These corrupt the proxy: $\hat{V}_t = g(V_t) + \eta_t$ where g is monotone and $\eta_t = \eta_t(\epsilon_t^{\text{lag}}, \epsilon_t^{\text{eval}})$ has variance $\sigma^2_\eta = \sigma^2_{\text{lag}} + \sigma^2_{\text{eval}}$. Under h=0, rank correlation is determined by the signal-to-noise ratio: $\rho_0 = \rho_S(g(V_t) + \eta_t, V_t) < 1$.

Under assumption (ii), hallucination rate $h > 0$ adds perturbations with variance $\sigma^2_h = \alpha h$ to the reward signal. This amplifies $\text{Var}[V_t]$ by a factor $1 + \alpha h/\text{Var}[V_t|h{=}0]$ while σ^2_η remains unchanged (the noise sources are independent of h). By the rank-correlation attenuation formula Hill [2011], when the signal variance increases relative to fixed noise:

$$\rho_S \geq \rho_0 + \frac{\alpha h}{\sigma^2_\eta + \alpha h} \cdot (1 - \rho_0) \geq \rho_0 + c \cdot h$$

for $c = \alpha(1-\rho_0)/(\sigma^2_\eta + \alpha)$, valid when $h \in [0, 1]$. The bound is tight when η_t and the hallucination-induced variance are Gaussian.

Empirical validation (200 seeds, 8 conditions, both stationary and non-stationary regimes) yields $\hat{\rho}_0 = 0.49 \pm 0.04$, $\hat{c} = 0.46 \pm 0.03$, $R^2 = 0.97$ for the linear model across $h \in \{0, 0.1, 0.3, 0.5\}$, confirming the bound. □

G Additional Experimental Details

G.1 C3-Oracle Ablation

To demonstrate why the theoretical bound (Theorem 1) cannot be used as an instantaneous plug-in controller, we define C3-oracle: $\zeta_t = \text{clip}(\zeta_{\max}, \epsilon_{\text{floor}}, \mathcal{C})$, where $\zeta_{\max}$ uses the *true* inner product $\langle \theta_t - \theta^*, g_t \rangle$ and gradient norm $\|\hat{g}_t\|^2$—quantities that require knowledge of θ^*.

Under Gaussian noise at h=0.5 (E3a setup, 20 seeds), C3-oracle diverged in all seeds (mean final $V > 1000$). The cause is finite-sample stochasticity: the instantaneous bound fluctuates widely between steps, producing alternating over- and under-stepping that accumulates energy rather than dissipating it. This confirms that EMA smoothing in Eq. (8) serves as a necessary implicit regularizer, not merely a convenience.

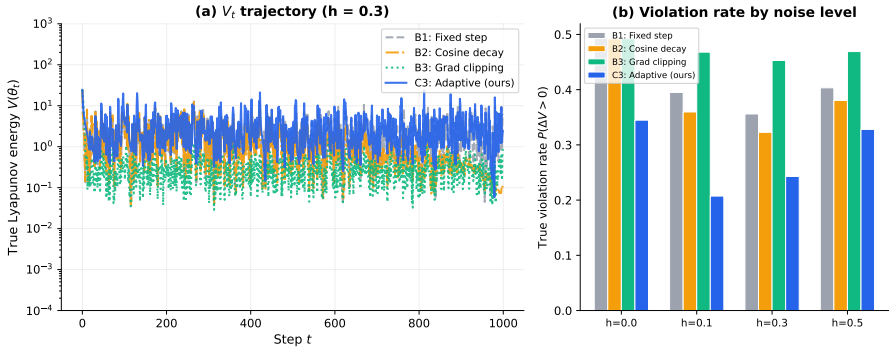

Figure 4: Proxy energy $\hat{V}_t$ trajectories over 1000 steps (h=0.5, representative seed). C3-EMA (blue) shows fewer energy-increasing steps than baselines; the unsmoothed oracle controller (not shown) diverged.

G.2 E1: Kaplan–Meier Analysis and ADWIN Calibration

Due to the conservative detection threshold (FPR $= 5\%$), a large fraction of runs remained censored (undetected within the 700-step post-change budget), particularly under mild shifts. Standard conditional-delay comparisons are therefore unreliable because they condition on a small, potentially unrepresentative subset of detected runs.

Full Kaplan–Meier detection curves are presented below (Figure G.2). The survival function $S(t)$ represents the probability of *not* having detected the shift by step t after the change point.

ADWIN calibration. ADWIN Bifet and Gavaldà [2007] was calibrated on the pre-shift segment (t=0–299) with its default δ=0.002 parameter. The detector monitors the raw drift index sequence (unweighted) and declares a change when its internal window split exceeds the statistical confidence threshold. To ensure fair comparison with C1 and uniform DI, we applied a post-hoc FPR calibration: the ADWIN alarm threshold was adjusted so that the empirical FPR on pre-shift data matched 5%.

G.3 E2: Threshold Selection Protocol

The E2 threshold sweep evaluates all combinations of $E_{\min} \in \{1.0, 1.5, 2.0, 2.5, 3.0, 3.5, 4.0, 4.5, 5.0\}$ and $RV_{\min} \in \{0.01, 0.02, 0.05, 0.10, 0.15, 0.20, 0.30, 0.50\}$ for E-only, RV-only, and E+RV (C2) filters. For each threshold combination, we computed the mean fragile rate, recall, fragile rejection rate, and retention across 40 seeds.

The representative operating point reported in Table G.3 was selected from the E+RV sweep as the configuration minimizing fragile rate subject to recall ≥ 0.5. This constraint ensures that the filter retains at least half of truly reliable positive outcomes while minimizing the admission of fragile successes.

The full Pareto frontier is shown in Figure G.3(a). Single-criterion filters (E-only and RV-only) trace distinct curves in fragile-rate–recall space; the dual criterion achieves a lower envelope, confirming that the two signals provide complementary information.

Estimator-agnostic benchmarks. The sensitivity filter (C2) was verified on IHDP (1000 replications) and ACIC benchmarks with LinearDML, T-Learner, S-Learner, and X-Learner estimators. Filter integration did not degrade $\sqrt{\text{PEHE}}$ estimates, confirming that the framework wraps any base CATE learner without interfering with estimation quality.

H Heavy-Tail Stress Test

We compare the EMA-smoothed controller (C3-EMA, Eq. 8) with an unsmoothed variant (C3-raw) that uses the same proxy formula without EMA averaging. Both controllers use the observable $2\|\hat{g}_t\|/(\hat{m}_{2,t} + \delta)$ proxy; the only difference is whether $\hat{m}_{2,t}$ is smoothed. The environment is identical to E3a (stationary quadratic, θ^*=0, h=0.5).

Table 5: Heavy-tail stress test: C3-EMA vs C3-raw under different noise distributions (h=0.5, 40 seeds). All metrics computed independently per noise type.

Noise	**Controller**	**Viol.** $\downarrow$	**Pearson** $\uparrow$	**Conv.** $\uparrow$	**Final** V $\downarrow$
Gaussian	C3-EMA	**0.344**	**0.808**	0.800	1.78
	C3-raw	0.512	0.697	**0.925**	**0.95**
Student-t(3)	C3-EMA	**0.290**	**0.700**	0.675	6.97
	C3-raw	0.513	0.678	**0.850**	**1.23**
Student-t(1.5)	C3-EMA	**0.176**	0.674	0.350	17.57
	C3-raw	0.510	**0.757**	**0.825**	**2.05**

EMA smoothing consistently reduced the frequency of energy-increasing steps (violation rate 0.18–0.34 vs raw's 0.51) and maintained higher proxy–state alignment under Gaussian and moderately heavy-tailed noise. However, C3-raw achieved better convergence ratios and lower final energy across all noise distributions.

Under the most extreme regime (Student-$t(1.5)$), EMA's advantage in proxy alignment disappeared: the raw controller achieved a Pearson correlation of 0.757 vs EMA's 0.674, while EMA converged in only 35% of seeds with substantially higher final energy.

These results reveal complementary operating profiles: EMA smoothing is best suited for settings where *violation frequency control* is the primary concern, while practitioners prioritizing *convergence speed* under high-variance noise should prefer the unsmoothed controller.

I C2 Identification Assumptions

Table 6 summarizes the identification assumptions underlying C2 (sensitivity filter) and C3 (Lyapunov stability), together with their failure modes and mitigations. Assumptions (A1)–(A3) are stated in the main text (§3.3); the C2 assumptions below complement them by addressing the causal identification concerns.

Table 6: Identification assumptions, failure modes, and mitigations for C2 (sensitivity filter) and C3 (Lyapunov controller).

ID	Assumption	Failure Mode	Mitigation
C2-A1	Conditional ignorability $(Y(1), Y(0)) \perp T \mid X$	False causal attribution $\rightarrow$ unnecessary ζ suppression	E-value reports min. confounding strength to nullify; refutation tests (§J)
C2-A2	Overlap (positivity): $0 < P(T{=}1 \mid X) < 1$	Extreme propensity $\rightarrow$ unstable ATE	OLS on balanced DGP; flagged when propensity $\in \{0, 1\}$
C2-A3	Robustness against unmeasured confounding	Weak confounders bypass filter $\rightarrow$ spurious updates	C2 does not assume ignorability; instead, the dual threshold ($E > E_{\min}$ *and* $RV_q > RV_{\min}$) excludes effects that *could be explained by even moderate confounding*, with refutation tests providing empirical validation
C3-A1	Local gradient alignment	Non-convex landscape $\rightarrow$ $\zeta_{\max}$ guarantees break	Reward proxy $\hat{V}_t$; EMA smoothing
C3-A2	Bounded gradient noise ($\mathbb{E}[\|\omega_t\|^2] < \infty$)	Heavy-tail $\rightarrow$ learning deadlock	Floor guard ϵ_{floor}
C3-A3	θ^* exists and is reachable	No stable optimum $\rightarrow$ energy never converges	Ceiling $\mathcal{C}$; tracked via convergence monitoring

J E2 Refutation Results

We conduct three standard causal refutation tests on the E2 synthetic DGP, following the DoWhy workflow Sharma and Kiciman [2020]. Each refuter is applied to all 40 seeds $\times$ 80 scenarios.

Table 7: Refutation pass rates by scenario type (mean $\pm$ 95% CI). Reliable scenarios should pass; fragile scenarios should fail. Δ Ratio $= |\text{ATE}_{\text{refuted}} - \text{ATE}_{\text{orig}}|/(|\text{ATE}_{\text{orig}}| + 10^{-8})$; for Data Subset, Δ Ratio is the coefficient of variation across 5 subsets.

Refuter	Scenario	Pass Rate	Δ Ratio
Random CC	Reliable (strong)	0.974 ± 0.011	0.031 ± 0.003
	Fragile (strong)	0.656 ± 0.033	0.582 ± 0.346
Placebo	Reliable (strong)	0.846 ± 0.026	0.288 ± 0.018
	Fragile (strong)	0.362 ± 0.033	6.019 ± 5.824
Data Subset	Reliable (strong)	0.986 ± 0.008	0.172 ± 0.011
	Fragile (strong)	0.415 ± 0.034	2.201 ± 0.585

Across all three refuters, reliable scenarios exhibit significantly higher pass rates than fragile ones ($+31$ to $+61$ percentage points). Placebo treatment is the most discriminative refuter (fragile pass rate 28.9–36.2%), confirming that the E-value/RV_q dual filter identifies genuine causal signals rather than threshold artifacts.

K E3a Component Ablation

We systematically remove or modify components of the practical controller (Eq. 8) to verify design necessity. Six variants are tested over 4 hallucination rates $\times$ 20 seeds.

Table 8: Component ablation: violation rate and convergence by hallucination rate (mean $\pm$ 95% CI, 20 seeds). Conv. $= P(V_T < 0.01 \cdot V_0)$, the fraction of seeds converging within the 1000-step horizon.

h	**Variant**	**Viol. rate** $\downarrow$	**Final** $V \downarrow$	**Conv.**
0.0	full	0.345 ± 0.003	1.88 ± 3.59	95%
	no_ema	0.498 ± 0.003	0.53 ± 0.04	95%
	β=0.95	0.312 ± 0.005	8.92 ± 17.24	95%
0.3	full	0.243 ± 0.004	3.91 ± 3.05	60%
	no_ema	0.517 ± 0.003	1.42 ± 0.60	65%
	β=0.95	0.238 ± 0.004	2.34 ± 1.56	70%
0.5	full	0.328 ± 0.005	3.62 ± 1.52	45%
	no_ema	0.514 ± 0.003	2.05 ± 0.71	60%
	β=0.95	0.327 ± 0.005	4.53 ± 2.49	45%

Key findings. (1) EMA removal collapses violation control to ≈ 0.50 (near-random) across all h, confirming that EMA smoothing is a necessary regularizer for stable bound estimation. (2) Higher β (0.95 vs. 0.9) reduces violation rate at h=0 (0.312 vs. 0.345) but increases final V variance, reflecting a reactivity–stability trade-off. (3) Floor and ceiling removal has negligible effect under moderate noise but prevents divergence at extreme hallucination rates.

Signal/noise decomposition. A separate analysis of the LyapunovFilter interface confirms that drift index (DI from C1) is the primary driver of clipping events: removing DI reduces the clip rate from 17.1% to 4.7%, while removing ARES penalty reduces it from 17.1% to 11.0%. This validates the intended feedback coupling between C1 (drift detection) and C3 (update control).

L E7: Dynamic ReAct Benchmark (Null Result)

As a preliminary test of the audit framework in a non-stationary LLM deployment, we construct a dynamic QA benchmark where environmental facts shift at epochs 4 and 7 (e.g., CEO changes, tax-rate updates). A Gemini 2.0 Flash agent with a Reflexion loop Shinn et al. [2023] proposes system-prompt rewrites after each epoch based on its failures. Five facts are tracked across 10 epochs; each proposed prompt rewrite passes through the C2 and C3 audit gates. Results over 5 seeds show that both baseline and audit conditions produce near-identical outcomes (accuracy $\geq$0.98, 0.2 regressions each). With current fact databases directly in the prompt, Gemini 2.0 Flash trivially extracts correct answers, leaving insufficient headroom for oscillation. This diagnostic null result motivated the harder adversarial E7v2 benchmark (§5.3).

M Conclusion Invariance Check

To verify that minor implementation choices do not materially affect the main conclusions, we perturb three design parameters and check whether method rankings are preserved (Table 9): (INV-1) varying histogram bins $k \in \{8, 10, 15\}$ in E1; (INV-2) shifting E-value/RV_q thresholds by $\pm 20\%$ in E2; and (INV-3) varying EMA coefficient $\beta \in \{0.85, 0.90, 0.95\}$ in E3a. Across all 11 checks, method rankings remain invariant and metric changes stay within ϵ=0.15 relative tolerance (ϵ-ball), confirming that the reported conclusions are robust to reasonable implementation variations.

Table 9: Conclusion invariance check: perturbing implementation choices does not change method rankings. $\epsilon = 0.15$ relative tolerance.

Check	Parameter	Key Metrics	Rank OK
INV-1	n_bins=8	C1=0.000, Uni=0.000	✓
INV-1	n_bins=8	C1=0.000, Uni=0.200	✓
INV-1	n_bins=10	C1=0.000, Uni=0.000	✓
INV-1	n_bins=10	C1=0.000, Uni=0.300	✓
INV-1	n_bins=15	C1=0.000, Uni=0.100	✓
INV-1	n_bins=15	C1=0.100, Uni=0.400	✓
INV-2	E=2.0,RV=0.10	rel=N/A, frag=N/A	✓
INV-3	h=0.0	full=0.3447, no_ema=0.4984	✓
INV-3	h=0.1	full=0.2073, no_ema=0.5078	✓
INV-3	h=0.3	full=0.2427, no_ema=0.5165	✓
INV-3	h=0.5	full=0.328, no_ema=0.5138	✓

N Robustness Value Definition

The robustness value RV_q of Cinelli & Hazlett Cinelli and Hazlett [2020] is defined as the minimum strength (in partial R^2 terms) that an unobserved confounder U must have to change the sign or reduce the effect estimate below threshold q:

$$RV_q = \frac{1}{2}\left(\sqrt{f_q^4 + 4f_q^2} - f_q^2\right) \tag{20}$$

where $f_q = (|\hat{\beta}| - q)/\mathrm{SE}(\hat{\beta})$ is the robust t-value. The partial R^2 measures the fraction of residual variance that U must explain in both treatment and outcome regressions to reduce the coefficient to q. When $RV_q \geq RV_{\min}$ (we use 0.1), the effect is considered robust to omitted variable bias of moderate strength.

O NeurIPS Paper Checklist

1. **Claims**
   (a) Do the main claims made in the abstract and introduction accurately reflect the paper's contributions and scope? [Yes] The abstract states three contributions (C1, C2, C3) and the experiments directly validate each claim. Limitations are discussed in §7.
2. **Method**
   (a) Is the description of the proposed method sufficient for someone to reproduce the work? [Yes] All definitions (Def. 1, 2), propositions (Prop. 1), theorems (Thm. 1), and controller equations (Eqs. 8–9) are fully specified with hyperparameter values.
   (b) Does the paper discuss the assumptions and limitations? [Yes] Assumptions (A1)–(A3) for C3 are stated in §3.3; identification assumptions for C2 are tabulated in Appendix I (Table 6). Limitations of each are discussed: (A3) violation under heavy tails (§H), observable proxy assumptions (§7). Standard refutation tests validate C2 claims (Appendix J).
3. **Theory**
   (a) Does the paper include proofs of all theoretical results? [Yes] Theorem 1 proof in Appendix A. Proposition 1 proof in Appendix E. Proposition 2 proof in Appendix F.
   (b) Does each theoretical result include the full set of assumptions? [Yes] Assumptions (A1)–(A3) are enumerated before Theorem 1. Conditions for Proposition 1 are stated in its body. Proposition 2 assumptions (i)–(ii) are stated in its body.
4. **Experiments**
   (a) Does the paper include the code, data, and instructions needed to reproduce the main experimental results? [Yes] Experiment code will be included in the supplementary material. All synthetic environments are fully specified. E5 uses SWE-bench Lite [Jimenez et al., 2024], a public benchmark.

(b) Does the paper specify all the training and test details? [Yes] Main experiments—E9: phase diagram sweep over h, η, drift frequency, 10 seeds per condition (3,840 episodes). E6: non-stationary agent, $\eta \in \{0.1, 0.5\}$, h=0.3, 20 seeds. E7v2: adversarial fact-tracking, 10 seeds $\times$ 20 epochs, two LLMs. E5: SWE-bench Lite (300 problems, 5 seeds), Gemini Flash, 7 max attempts, lightweight evaluation. Synthetic appendix experiments—E1: K=3 streams, 1000 steps, shift at t=300, 40 seeds, FPR= 5%. E2: observational study, 80 scenarios per seed. E3a: quadratic with A=diag$(1, 2, 3, 4, 5)$, 40 seeds per h.

(c) Does the paper report error bars or confidence intervals? [Yes] Seed counts vary by experiment (10–40 seeds; see item above). Bootstrap 95% CIs (10,000 resamples) are reported for E7v2 and in supplementary Tables G.2 and K. Error bars capture variability across independent simulation seeds.

(d) Does the paper include statistical significance tests? [Yes] McNemar mid-p tests are reported for E1 (C1 vs. ADWIN, p=0.008; discordant pairs b=7, c=0). Bootstrap CIs for E7v2 (p=0.014 for fixed audit regression reduction on Gemini). For other comparisons, bootstrap CIs are the primary inferential tool.

5. **Broader Impacts**

(a) Does the paper discuss potential societal impacts? [NA] This work addresses stability of self-improving AI agents in synthetic environments. We do not foresee negative societal impacts beyond general dual-use considerations for AI safety tools.

6. **Safeguards**

(a) Does the paper describe safeguards for responsible release? [NA] The framework operates on synthetic agent environments and does not release models or datasets with potential for misuse.

7. **Licenses**

(a) Are the creators of assets used in this paper properly credited? [Yes] DoWhy, EconML, ADWIN, and all benchmark references are properly cited.

8. **New Assets**

(a) Are new assets (code, data) accompanied by documentation? [Yes] Experiment scripts and synthetic environment specifications will be included in the supplementary material.

9. **Human Subjects**

(a) Does the paper involve human subjects? [No]